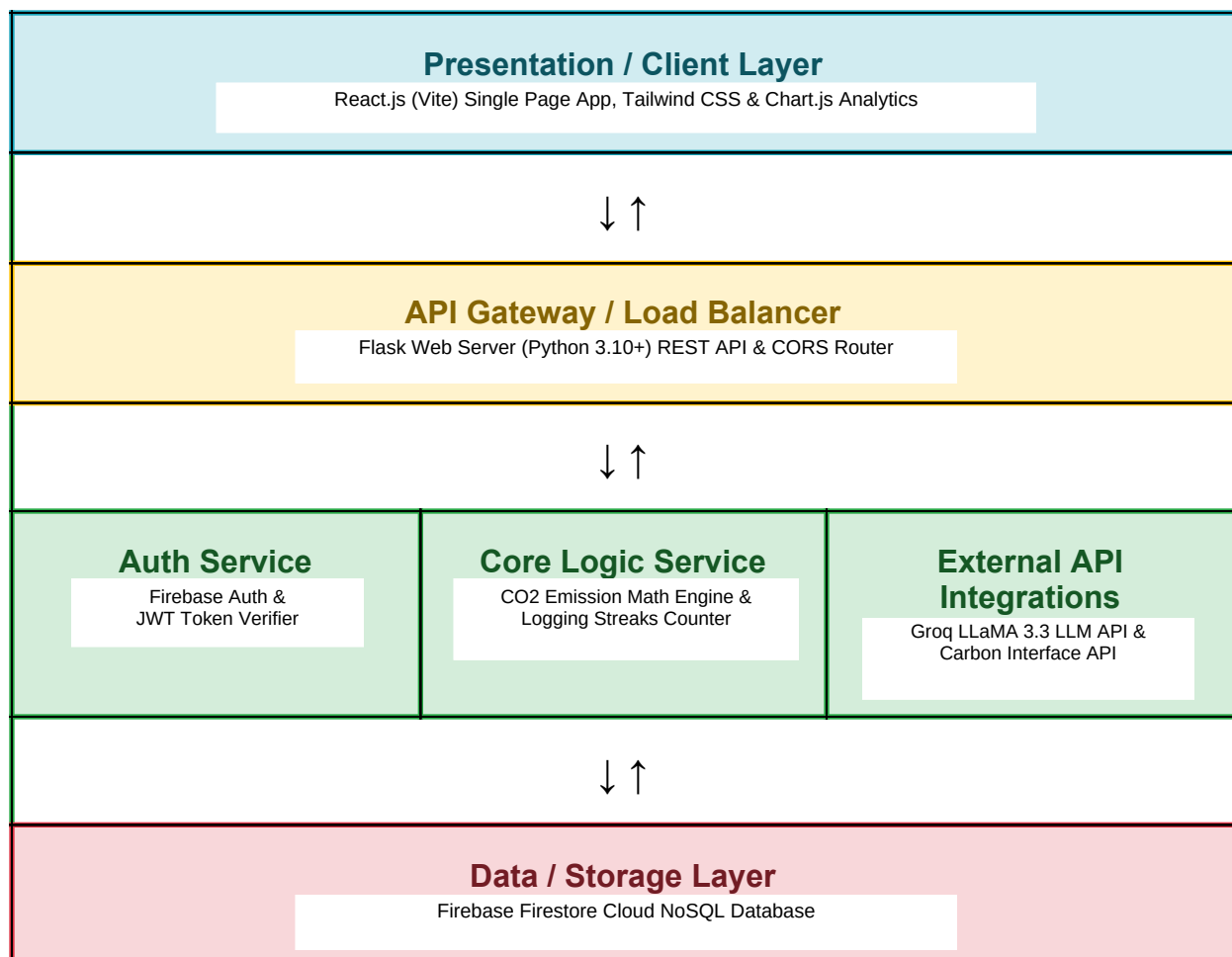


Solution Architecture

Date	15 June 2026
Team ID	
Project Name	EcoTrack - Personal Carbon AI Footprint Tracker
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders dynamic carbon dashboards, daily habit logging forms, 30-day Chart.js analytics, & AI swap cards.	React.js (Vite), Tailwind CSS, Chart.js
API Gateway & Server	Handles client HTTP requests, enforces CORS isolation, routes API endpoints, & manages environment variables.	Flask Web Server, Python 3.10+
Core AI & Logic	Calculates daily CO2 emissions math, computes logging streaks, and fetches Groq LLaMA 3.3 AI eco-swaps.	Groq LLM API, Carbon Interface API
Database & Storage	Persists user habit logs, carbon history, streak counts, & manages user authentication sessions.	Firebase Firestore, Firebase Auth